

Advanced (Carolina Reaper)

Q1. Simplify $\sqrt{8} + \sqrt{18}$

- (A) $4\sqrt{2} + 9\sqrt{2}$
- (B) $2\sqrt{2} + 3\sqrt{2}$
- (C) $6\sqrt{2}$
- (D) $4\sqrt{2}$
- (E) $9\sqrt{2}$

Q2. Simplify $\sqrt{32} \cdot \sqrt{2}$

- (A) 8
- (B) 4
- (C) 16
- (D) $\sqrt{64}$
- (E) $2\sqrt{16}$

Q3. $\sqrt{50} - \sqrt{18} =$

- (A) $\sqrt{32}$
- (B) $2\sqrt{2}$
- (C) $\sqrt{8}$
- (D) $5\sqrt{2} - 3\sqrt{2}$
- (E) 8

Q4. A carpenter needs to cut a $12\sqrt{3}$ feet long wood. If she already has a $4\sqrt{3}$ feet long piece, what length of wood does she still need to cut?

- (A) $8\sqrt{3}$ feet
- (B) 12 feet
- (C) 4 feet
- (D) $16\sqrt{3}$ feet
- (E) 20 feet

Q5. $\frac{\sqrt{24}}{\sqrt{6}} =$

- (A) $\sqrt{4}$
- (B) 2
- (C) $\sqrt{2}$
- (D) $\sqrt{3}$
- (E) 4

Q6. A furniture maker has to build a rectangular table with an area of $48\sqrt{2}$ square feet. If the length is $4\sqrt{2}$ feet, what is the width?

- (A) 12 feet
- (B) 8 feet
- (C) 6 feet
- (D) $4\sqrt{2}$ feet
- (E) $3\sqrt{2}$ feet

Q7. A cyclist travels $12\sqrt{5}$ miles in 2 hours. What is her average speed in miles per hour?

- (A) 6 miles per hour
- (B) 12 miles per hour
- (C) 5 miles per hour
- (D) 8 miles per hour
- (E) $4\sqrt{5}$ miles per hour

Q8. $\sqrt{72} - \sqrt{12} =$

- (A) $6\sqrt{2} - 2\sqrt{3}$
- (B) $6 - 2\sqrt{3}$
- (C) 4
- (D) $2\sqrt{2} - 2\sqrt{3}$
- (E) $2\sqrt{3}$

Open Ended Questions (FRQ)

Q9. Water flows from a garden hose at $\sqrt{18}$ gallons per minute. How many minutes will it take to fill a 36-gallon bucket?

Q10. A surfer is $4\sqrt{2}$ miles away from the shore. If she can surf at an average speed of $2\sqrt{3}$ miles per hour, how many hours will it take her to reach the shore?

Chef's Tips

- Ensure students simplify their radicals.
- Emphasize the importance of the correct order of operations.
- Encourage students to check their units in word problems.